

Tutorial - 5

1. Consider a CMOS process flow with the following parameters: $V_{T_n} = 0.6V$, $\mu_n C_{ox} = 60\mu A/V^2$ and $V_{T_p} = -0.7V$ and $\mu_p C_{ox} = 25\mu A/V^2$. The power supply is 3.3V and the gate length is $0.8\mu m$.
 - Find the width ratio so that the switching threshold of the circuit is at 1.4V
 - The CMOS fabrication process used to manufacture this inverter allows a variation of the V_{T_n} value by $\pm 15\%$ around its nominal value, and a variation of the V_{T_p} value by $\pm 20\%$ around its nominal value. Assuming that all other parameters always retain their nominal values, find the upper and lower limits of the switching threshold voltage (V_{th}) of this circuit.
2. Consider a CMOS inverter with the same transistor parameters as in Q1, with $(W/L)_n = 8$ and $(W/L)_p = 12$. Calculate the noise margins and the switching threshold of the circuit. Assume that $V_{DD} = 3.3V$.
3. Sometimes to save area, the following circuit is used as an inverter. This logic style is known as pseudo-nMOS. (out of favor)
 - Assuming that $V_{T_n} = -V_{T_p}$ and $k_n = 5k_p$ find V_{OH} and V_{OL} .
 - For $k_n = 300\mu A/V^2$, $V_{T_n} = 0.4V$ and $V_{DD} = 1.8V$, evaluate the values of V_{OH} and V_{OL} . Find the power dissipated in the inverter if it spends half the time in each of its two states.
4. Consider a CMOS inverter with $V_{DD} = 1.8V$, $V_{T_n} = -V_{T_p} = 0.5V$, $\mu_n = 4\mu_p$ and $\mu_n C_{ox} = 300\mu A/V^2$. The NMOS is sized such that $L = 0.18\mu m$ and $(W/L)_n = 1.5$.
 - Find the W_p that results in the switching threshold to be at 0.9V. What is the silicon area utilised by the inverter.
 - For the matched case in the above section, find the 5 characteristic voltages and compute the noise margins. Compute the value of V_{out} when $V_{in}=V_{IL}$: this can be considered as the worst case value of V_{OH} . Similarly compute the worst case value of V_{OL} , by assuming $V_{in}=V_{IH}$. Use these worst case values to determine more conservative values for the noise margins.
 - For the case in part (i) find the output resistance of the inverter in both states
 - If $\lambda_n = |\lambda_p| = 0.2V^{-1}$, what is the gain of the inverter at the switching threshold? If a straight line is drawn through the point $V_{in}=V_{out}=V_{TH}$ with the slope equal to gain, at what values of input voltage does it intercept the vertical lines of $V_{out}=0$ and $V_{out}=V_{DD}$. Use these intercepts to estimate the width of the transition region of the VTC.
 - If $W_n=W_p$, what is the value of V_{TH} ? What do you estimate the reduction in NM_L (relative to the matched case) be? What is the percentage savings in Si area?
 - Repeat the above for the case when $W_p=2W_n$. This case is generally used in the industry. Why?

